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ARTP Standards of Care - CPAP Devices (Technical and Performance)

1.0 INTRODUCTION

The NICE. CPAP Health Technology Appraisal (HTA), published in March 2008, recommended CPAP as an effective technology treatment for Obstructive Sleep Apnoea/Hypopnoea Syndrome (OSAHS). The consequence of this HTA saw a vast expansion in OSAHS diagnostics, CPAP assessment and CPAP issue in the United Kingdom. The NICE guideline “*Obstructive sleep apnoea/hypopnoea syndrome and obesity hypoventilation syndrome in over 16s*” published in August 2021 reinforced the use of CPAP, both fixed pressure and auto-CPAP devices for the treatment of OSAHS in the over 16s.

EN ISO 80601-2-70:2020 (Medical electrical equipment - Part 2-70: Particular requirements for the basic safety and essential performance of sleep apnoea breathing therapy equipment) sets down standards for the manufacture and performance of CPAP devices. *EN ISO 80601-2-70:2020* does not legally compel manufacturers of CPAP devices to commit to produce devices which are guaranteed to 100% comply with that standard. *EN ISO 80601-2-70:2020*, as with all ISO protocols, also allows considerable leeway for manufacturers to deviate from what is prescribed in the ISO.

The risk to patients of unregulated provision of CPAP devices without regard to technical and performance quality standards is real and significant. There is the possibility of sub-optimal patient care being delivered, albeit unwittingly, due to ineffective and inappropriate treatment being delivered by the provision of poorly performing CPAP devices. Whilst most CPAP device manufacturers and distributors strive to provide quality products and service, there is no way of minimising the potential for poorly performing devices to be sold without a robust, independent testing regimen. Indeed,

the circumstances which led in the United Kingdom to the National Patient Safety Alert: *Philips ventilator, CPAP and BiPAP devices: Potential for patient harm due to inhalation of particles and volatile organic compounds (NatPSA/2021/005/MHRA)*, demonstrates clearly that rigorous independent testing regimes are required for CPAP devices and many other categories of medical equipment. This one incident has had a devastating effect on the procurement and supply of CPAP devices on a global basis, at a time of global pandemic, which will take some years for the situation to return to anything like normality. There is a potential danger that a global shortage of CPAP devices from reputable manufacturers allows for less reputable manufacturers, particularly from the Far East, to profit and to supply a raft of poor quality, deficient CPAP devices on a global basis with no regulation to curtail such activities.

Relevant minimum technical and performance standards for CPAP devices have existed in Germany for over a decade. GKV-Spitzenverband (GKV-S), as the central association of the health insurance funds at federal level, maintains the 'Hilfsmittelverzeichnis', the 'List of Aids', known within the medical device industry as the '*Red Book*'. Under Product Group 14 - Inhalation and Breathing Apparatus, Subgroups 20 and 21 refer to Fixed Pressure CPAP and Auto CPAP devices respectively. The 'Medizinische Anforderungen' - 'Medical Requirements' for Subgroups 20 and 21 identify the technical standards of performance required before CPAP devices can be placed upon the official 'List of Aids' available via the German Statutory Health Insurance System. The standards set out a recognised and agreed testing procedure for verification of compliance of any device with these standards in respect of operational noise level, stability of the static and dynamic airway pressure accuracy, gas purity, warming the inhaled air and, where appropriate, proof of humidification performance.

Under the German standards, technical standards of performance are documented as compliant either by submission of manufacturer in house testing data or independent laboratory testing.

The technical standards of performance for CPAP devices set out in this document require that independent laboratory testing is performed on a sample device to ensure compliance with requirements for pressure stability, noise, and humidifier performance. Whilst the standards in this document originally aligned with those required for the German '*Red Book*', amendments have been made to this standard to keep up to date with ISO changes which the '*Red Book*' does not. CPAP device manufacturers, and distributors where the manufacturer does not have a presence in the United Kingdom, who are members of the ARTP Sleep Apnoea Consortium (ARTP SAC), may submit CPAP devices to ARTP of their own volition for independent laboratory testing against the technical and performance standards set out in this document. ARTP will maintain and publish a register via its web site of those sampled devices that meet or exceed the technical and performance standards set out in this document. Where a submitted device does not meet the required standards of performance, a detailed report of the independent testing outcome will be shared with the submitting organisation. There is no intention by ARTP to publish details of submitted devices, which do not meet the required standards. Where manufacturers can provide test reports from independent

laboratories, these may be accepted as sufficient evidence for a device to meet the requirements for the register of acceptable devices.

The register of CPAP devices which have been tested in an *independent laboratory* commissioned by ARTP and which meet or exceed the ARTP technical and performance standards will serve as a reference to all providers of CPAP therapy that such devices have been verified to meet internationally recognised standards of performance and which are therefore satisfactory to provide to patients.

The standards in this document provide testing methodology for devices in anything fixed pressure mode for pressure stability, noise, and humidifier performance. A draft methodology and expected outcomes for auto adjusting (commonly known as auto-titrating and referred to using the acronym APAP) CPAP devices are presented in this version of this standards document. ARTP will work with interested professionals and organisations internationally to develop standards specifically relating to the additional performance features of APAP devices.

CPAP devices have several features that are used to market the devices but clearly there are certain features that are essential to providing an effective device and those additional that are desirable. These features are discussed below.

1.1 Essential Features of CPAP Devices

These are basic features that comprise the minimum standard expected from a CPAP device and aim to cover those features required to deliver good device performance to the patient and enable tolerance and compliance with the therapy.

1.2 Desirable Features of CPAP Devices

These are often manufacturer specific features that add extra benefit to patients or service providers that may improve patient compliance, comfort, and tolerance of the treatment. There may not be clear independent evidence that they make any difference in randomised controlled trials, but they offer potential benefits and an extended range of equipment to allow tailored devices for specific patients.

ESSENTIAL FEATURES

2.0 EQUIPMENT BASICS

2.1 General Considerations

2.1.1 International/European Standards

CPAP devices must comply with International/European standards which include:

- *IEC 60601-1: General Requirements for Safety of Medical Electrical Equipment*
- *EN ISO 80601-2-70:2020: Particular requirements for basic safety and essential performance of sleep apnoea breathing therapy equipment.*
- CE Marking is essential.

2.1.2 Independent Laboratory Testing

Evidence from Independent Laboratory Testing will be accepted from manufacturers where ARTP is satisfied that the testing procedures are truly independent of manufacturer influence and ARTP has been able to satisfy itself that the testing facilities are *bona fide* and perform testing to the required standards.

2.2 Operational Modes

There must be two modes of operation as detailed in 2.2.1 and 2.2.2.

2.2.1 Prescriber/Clinician Mode - enables settings to be made and adjusted by the prescriber. The prescriber can lock the settings made so that the user cannot adjust the parameters set.

2.2.2 User Mode - enables a limited range of comfort settings to be made and adjusted by the user.

2.3 Operating Instructions

Operating instructions for both prescriber and user must be in the English language with appropriate clear diagrams, figures, and pictures. CPAP devices and accessories must be easy to operate and use.

2.4 Cleaning/Infection Control

CPAP devices and accessories must be designed to be easy to keep clean using widely available domestic cleansing agents. User manuals must contain precise cleaning/disinfection instructions.

CPAP devices must be constructed in such a manner so that there are no 'crevices' which are difficult to clean. For normal day to day cleaning, CPAP devices should be able to withstand surface cleaning with a cloth dampened with a water and mild detergent combination or, alternatively, commercially available surface disinfectant wipes.

User cleaning of masks, straps, tubing, and any other accessories must be achievable by

soaking in a warm water and mild detergent solution with a period of agitation followed by thorough rinsing in clean water and air drying.

If a humidifier is used it must be possible to completely dismantle the humidifier and disinfect all parts thermally in a domestic dishwasher. The manufacturer must specify in the user manual what temperature limit and specific dishwasher programme temperature (e.g., 65°C, 55°C, Delicate/Glass, etc.) and shelf placement requirements apply. Instructions to prevent and remove build-up of mineral deposits from the humidifier using domestically available reagents (e.g., a dilute solution of vinegar) must be provided.

Recently, consumer devices, which are readily available, purporting to “eliminate 99.9% of CPAP bacteria, germs, and other pathogens safely and naturally with no water or messy chemicals needed”, and also claiming “compatibility with major CPAP models” have come into being. The use of such devices is not recommended by any of the major CPAP device manufacturers. Using such devices may invalidate warranties for CPAP Devices, CPAP masks, headgear, and tubing accessories. A review of the available literature yields no evidence that the risk of infection from the use of traditional cleaning methods, as recommended by manufacturers, is significant. ^[25,26]

2.5 Dimensions

Weight must not be more than 1.5 kg for the non-humidified CPAP unit and less than 2.0kg with an integral humidifier (without water). The weight specified includes that of any external power supply.

Size: Sum of the 3 dimensions, Width (W), Depth (D) and Height (H) should be no greater than 48 cm including wireless/Bluetooth add-on module(s). This is to allow for easy transport of the CPAP when travelling on aeroplanes etc. With an integrated humidifier attached, the sum of W+D+H must not increase by more than 50% of the original size (W+D+H) of the stand-alone device; thus, the upper limit for size is 72 cm.

2.6 Connectors

These must be 15 or 22 mm, conical, and comply with the requirements of the current applicable version of *EN ISO 80601-2-70:2020*.

2.7 Tubing

The device must be supplied with robust but flexible tubing, nominally 1.8m length, which fits firmly but easily and must comply with the requirements of the current applicable version of *EN ISO 80601-2-70:2020*. The original or 'standard' diameter for a hose used with CPAP therapy was 22mm and these can still be used with most devices. However, certain devices require 15mm tubing and are designed to work only with a 15mm hose. Some devices allow for the selection of either a 22mm or a 15mm hose. There are yet other devices using only a heated hose specific to that line of devices and which cannot use a non-heated hose of any diameter. The tubing must be able to be cleansed using recommended and widely available household cleansing agents and be able to withstand disinfection to recognised hospital standards (low temperature steam at 90°C, ethylene oxide).

2.8 Size, Shape and Form of Buttons/Controls on Device

Control buttons must be large enough so that they are easily found in dark conditions. The use of raised patterns (like Braille) on control buttons is encouraged. The use of controls which illuminate for a short period when pressed is encouraged.

3.0 PRESSURE MEASUREMENT, STABILITY AND RELATED ASPECTS

3.1 Units of Pressure

The appropriate SI unit of pressure measurement for a CPAP device is **Hectopascals (hPa)**. This has been determined by national legislation (Weights and Measures Act 1985, amended 1994.) and EU directive (1995). Advice to the National Health Service at that time indicated that documenting non-SI units in patient case records could lead in the future to legal challenge and that therefore use of SI units was obligatory.

[Note: for practical purposes the units hPa, cm H₂O and mbar are numerically equivalent. i.e., 1 hPa = 1 mbar = 1 cm H₂O]

3.2 Standardisation of Pressure Units

In the interests of standardisation, the recommended unit of pressure measurement for CPAP devices in both the device and any applicable software supplied by the manufacturer/vendor should be the Hectopascal (hPa). Use of cmH₂O or mbar will continue to be permissible; no other units of measurement shall be used.

3.3 Pressure Range

Devices must deliver pressures from a **minimum of 2-4 hPa/cmH₂O** to a **maximum of 20 hPa/cmH₂O** (in not greater than 0.5 hPa/cmH₂O increments).

3.5 Pressure Stability (Long Term Accuracy)

The mean measured static pressure of CPAP devices must be within ± 0.5 hPa/cmH₂O for set pressures of <10.0 hPa/cmH₂O and ± 1.0 hPa/cmH₂O for set pressures of ≥ 10.0 to 20.0 hPa/cmH₂O unless the manufacturer specifies a lower tolerance in the technical specification for the device. The test method is as specified in *ISO 80601-2-70:2020*.

Pressure checks should be performed by providers of service using a manometer that is routinely checked against a standard (e.g., water manometer) every 6 months across its range. It is advisable to allow a stabilisation period of at least 20 minutes before attempting to measure pressures to allow for device warm up and algorithms relating to features such as altitude adjustment to take effect. Any deviation of measured pressure from set pressure during the first 20 minutes of operation from device activation may be due to such factors.

3.6 Pressure Stability (Short Term Accuracy)

The short-term accuracy (i.e., pressure variability peak to peak) must be less than 1.0 hPa/cmH₂O regardless of the set pressure at which the test is performed. The measurements must be made in accordance with *EN ISO 80601-2-70:2020* at minimum (P_{min}), $P_{min} + \frac{1}{4} (P_{max}-P_{min})$, $P_{min} + \frac{1}{2} (P_{max}-P_{min})$, $P_{min} + \frac{3}{4} (P_{max}-P_{min})$, and

maximum (P_{max}) pressure with breaths per minute (BPM) set to 10, 15, and 20 BPM.

The short-term accuracy for pressure stability must be ascertained by independent laboratory testing according to the protocol specified in Appendix B. The degree of uncertainty of the measurement should be stated by the independent testing laboratory and should not be more than $\pm 5\%$. In determining whether a tested device passes this test, the degree of uncertainty should be considered.

3.7 Leak

Unintentional leak compensation capabilities must allow continued safe operation of the CPAP device and the control algorithms and flow capabilities must allow a maximum flow capability greater than 30 LPM.

3.8 Adjustable Delay/Ramp

The ramp minimum pressure is a prescriber setting and should be adjustable from 4 hPa/cmH₂O in no greater than 0.5 hPa/cmH₂O increments. The delay/ramp period should be adjustable from Zero to 45 minutes maximum with the facility for the prescriber to set and lock in a shorter maximum period or disable the feature entirely. The use of algorithmic techniques which detect sleep/wakefulness and vary the ramp pressure from a prescriber pre-set level up to therapeutic pressure over a variable period of time is permitted where there is peer reviewed evidence in support of the technique. Users should be able to initiate the ramp feature at their discretion unless the prescriber has determined otherwise for valid clinical reasons.

3.10 Compliance Meter ("at pressure")

The compliance meter should not just measure motor usage but should measure hours run at pressure. Some devices can detect when breathing is occurring, and that the device is not just on with a fixed resistance. This is important to ensure that patients are complying with treatment and may have medico-legal implications for CPAP patients involved in road traffic accidents, LGV/PSV drivers, and other professional occupations such as airline pilots, train drivers, surgeons, high court judges, barristers, ship's captains, etc.

3.11 Leak Compensation

The device must automatically compensate for unintentional leaks.

3.12 Pressure Display

The actual pressure should be displayed rather than a permanent fixed/set value. Patients gain more confidence from seeing an actual pressure display and learn to 'trust' the device.

3.13 Maximum Limited Pressure Protection During Fault Condition(s)

A protection device to prevent the airway pressure from exceeding the maximum limited pressure of 30 hPa in normal conditions and of 40 hPa in a single fault condition shall be provided in accordance with *EN ISO 80601-2-70:2020*.

3.14 Altitude Operational Range and Altitude Compensation

The device must be capable of performing correctly over a minimum altitude range of sea level to 2500 metres. Automatic compensation for altitude is mandatory.

4.0 NOISE - ACCEPTABLE RANGE

Noise levels can often influence tolerance by the patient (or their partner) i of long-term use of the CPAP. The acceptable noise range (Sound Pressure Level) will be <28dB at 10 hPa/cmH₂O at a distance of 1 metre (independent, standardised assessment conditions). Manufacturers in their technical specifications must quote noise measured according to *EN ISO 80601-2-70:2020*. Conducted noise can be quoted in addition.

Noise must be ascertained by independent laboratory testing according to the protocol specified in Appendix A. The use of sound baffles/silencers in-line with the patient tubing to reduce noise is not permitted. These are a substitute for inferior blower design. Such inserts often cannot be situated in the delivery circuit before a humidifier and may become a source of occlusion as the insert becomes wet. We recommend that if any noise abatement materials are used in devices that they are materials tested to ensure they are not harmful to patients when used clinically (including testing any cleaning agents on those materials).

5.0 POWER SUPPLIES AND POWER CONSUMPTION

5.1 Power Supply

All CPAP devices must have an automatic universal power supply that will allow 100 to 240 V AC operation at 50-60Hz with no need for changes to any settings. Options for 12V or 24V DC operation are encouraged.

5.2 Power Consumption (Without Humidifier)

Actual power consumption depends on several factors such as height above sea level, ambient temperature and the accessories being used. Maximum power consumption without humidifier on 230 V AC power must be < 40 watts.

6.0 SETUP OF PRESCRIBED PARAMETERS

This must be able to be achieved either directly from the operational keys on the CPAP device or using PC/Mac software by means of remote control, data card or similar technologies. A default option for the settings applied by the prescriber to be locked and non-adjustable by the user is mandatory.

7.0 HUMIDIFICATION

7.1 The Relationship between Air Temperature and Relative Humidity

The amount of water vapor or humidity contained in air depends on the barometric pressure and temperature. **Absolute Humidity (AH)** is the actual or absolute amount of water vapor present in any given volume of air expressed in mg H₂O/L. The maximum amount of water that can be held at a given temperature and pressure, also expressed in mg H₂O/L, is the **Maximum Capacity**. **Relative Humidity (RH)** is the ratio of the AH to the maximum capacity, expressed as a percentage. Increasing temperature increases the capacity of a gas to hold water vapour and therefore, unless AH also increases, a higher temperature decreases RH and vice versa. Air is saturated or at the **Dew Point** temperature when the RH is 100%; i.e., when the water content equals the maximum capacity. Any subsequent decrease in temperature below the dew point results in condensation. This becomes extremely important when considering the effectiveness of humidifiers and the delivery of warm moist air in a cool environment and/or through a cool tube. Some devices offer an alternative heated patient delivery tube which enables a higher delivery temperature to be achieved as well as the potential for reduced condensation.

The relationship between temperature and relative humidity is shown in the table below:

Air Temperature °C	0	5	10	15	20	25	30	35	40
Dew Point Temperature °C	Relative Humidity (%RH)								
0	100	70	50	36	26	19	14	11	8
5	-	100	71	51	37	28	21	16	12
10	-	-	100	72	53	39	29	23	17
15	-	-	-	100	73	54	40	30	23
20	-	-	-	-	100	74	55	42	32
25	-	-	-	-	-	100	75	58	34
30	-	-	-	-	-	-	100	76	46
35	-	-	-	-	-	-	-	100	60
40	-	-	-	-	-	-	-	-	100

7.2 Physiological Humidification

Physiological humidification temperatures (i.e., 37°C at the lungs) are too hot for a CPAP device end of hose temperature. As CPAP does not bypass the upper airway, a lower temperature which considers the desired humidity for the patient for comfort and the additional patient contribution of heat and humidity into the system (both via expired air and air inspired and travelling through the upper airways) is more desirable. RH is more important to patient preference and symptoms than end of hose mask temperature, or AH^[19].

7.3 Humidifier Type and Performance

When addition of humidification is deemed appropriate CPAP devices should aim to deliver a level of RH which allows a range of options for patients to select, depending on

their preference, between cold/dry (where RH is lower) and warm/moist air (where RH is higher). Thus, any device should include a range of values that consider the patient contribution of humidity to the system, to deliver a consistent RH for the patient in changing conditions while preventing, as much as is practicable, condensation build up in the delivery tube.

There are now in existence a range of types of humidifier with proprietary functions such as 'Adaptive Humidification', 'Pulse Dose Humidification', etc. aimed at providing appropriate humidification and comfort whilst reducing the possibility for condensation or 'rain out' as it is often referred to. Previous versions of this standard have been aimed specifically at standard heated humidification without a heated patient circuit. The range of temperatures delivered by such devices at end of hose is lower than that achieved by devices with heated delivery tubes. Consequently, it is unreasonable to expect and, indeed, undesirable for devices with heated delivery tubes to achieve the same RH range as that required of standard heated humidifiers.

7.4 Standard Heated Humidifier - Expected Performance

A standard heated humidifier must be able to heat supplied air from room temperature to an end of patient hose temperature (using a standard 180 cm patient tube) achieving a temperature above at least 25°C in a number of transitional steps. It is unlikely that such a configured device will achieve an end of hose temperature of > 30°C. Adequate humidification must be ensured so that relative humidity can be maintained above at least 70% to 90%; this range being appropriate for the expected range of end of hose temperature which can be achieved. *Humidifier performance must be ascertained by independent laboratory testing according to the protocol specified in Appendix C.*

7.5 Humidifier with Heated Delivery Tube - Expected Performance

A humidifier with a heated delivery tube must be able to heat supplied air from room temperature to an end of patient hose temperature (using a standard 180 cm patient tube) achieving a temperature above at least 25°C in a number of transitional steps. It is probable that such a configured device may achieve an end of hose temperature of > 30°C. Adequate humidification must be ensured so that relative humidity can be maintained above at least 60% to 90%; this range being appropriate for the expected range of end of hose temperatures which can be achieved. Humidifiers in this category should not achieve end of patient tube temperature in excess of 33°C. *Humidifier performance must be ascertained by independent laboratory testing according to the protocol specified in Appendix C.*

7.6 Humidifier Connection to CPAP Device

Humidification capabilities should be an integrated feature of the CPAP device or be so constructed as to allow the addition of a dedicated humidification chamber to form a combined unit at a later date without the need to change the patient's CPAP device.

7.7 Maintenance

A humidifier, like a CPAP device, must comply with all relevant standards (CE). A humidifier must be easily removable by the user, cleansable with common recommended household cleansing agents and dishwasher safe.

7.8 Power Consumption with Humidifier Operational

Actual power consumption depends on several factors such as height above sea level, ambient temperature and the accessories being used. Maximum power consumption with a humidifier operational on 230 V AC power must be <100 watts regardless of whether a heated delivery tube is used or not.

7.9 Safety

Humidifier safety – when used with heated humidification, the device should provide water ingress protection if tilted 90 degrees in any direction.

8.0 SERVICE

CPAP devices and humidifiers must be supplied complete with documentation specifying the manufacturer's recommended requirement for acceptance testing, on-going electrical safety checks, and routine servicing (including who is deemed to be competent to perform these tasks). Devices should have minimal requirements for preventative maintenance in addition to filter changes.

It is acceptable for manufacturers to provide a statement of recommendation with a CPAP device indicating that this Class II, double insulate device does not require annual electrical safety checking. Likewise, it is acceptable for manufacturers to provide a statement of recommendation with a CPAP device indicating that it requires no user servicing for the life of the device apart from routine air filter cleansing/replacement.

8.1 Indication of Fault Conditions

The device must indicate conditions that the user can rectify via the device display. For all other fault conditions which require expert or manufacturer intervention, the device must indicate via the device display that a fault exists to the user, give a fault code, and advise the user to refer to the provider/manufacturer.

9.0 WARRANTY

The minimum acceptable manufacturer/vendor warranty period for parts and labour is 2 years. Manufacturers/vendors should provide an option to extend the warranty period from two years with annual increment options up to five years by payment of an additional sum at the time of purchase. Manufacturers/vendors are entitled to provide additional warranty periods above two years free of charge. This should be an entitlement for all purchasers within the United Kingdom from any specific manufacturer/vendor that should be made known to all purchasers and not be restricted to specific customers as and when determined by that manufacturer/vendor. Manufacturers/vendors must state what is included within their warranty conditions in order that customers can fully compare like with like.

10.0 AUTO-ADJUSTING CPAP DEVICES

Auto-adjusting CPAP devices are designed to increase pressure as needed to maintain airway patency and then to decrease pressure if no events are detected over a set period. Because the minimum effective pressure is delivered (auto-adjusting), the mean pressure is often lower than the optimal fixed CPAP pressure. There is currently no international standard relating to the performance of auto-adjusting devices. Auto-adjusting CPAP devices are commonly referred to as APAP devices. The Acronym APAP is defined as **Automatic Positive Airway Pressure**.

Currently, a wide variety of APAP devices are available. These devices vary in which physiological variables they monitor (snoring, apnoea, hypopnoea, airflow limitation, impedance, or a combination of variables) and in the algorithms used to respond to these factors. It is not the purpose of this standard of care document to specify which physiological variables and algorithms (usually proprietary and patented) are valid. What matters is that any auto-adjusting device can identify and react appropriately, in a timely manner, to obstructive apnoea and hypopnoea events.

Laboratory testing of APAP devices has been shown to have clinical relevance. [22] Clinical studies have shown that limitation in efficacy may be due to a failure of APAP devices to reliably estimate the therapeutic pressure in up to 25% of patients. [11,12,13,14] Previous studies where devices have been bench tested have predicted detection and response limitations under simulated obstructive event conditions. [15, 16, 17, 18]

10.1 Recognition of Respiratory Events

It is neither feasible nor desirable for any auto-adjusting device to recognise, classify and react to every respiratory event arising. Recognition of a high percentage of events arising is an essential feature and a goal that manufacturers must achieve. However, if a device was to react to every respiratory event, then the resultant continuous pressure adjustment, both increase and decrease, may not be beneficial to the patient and may produce a deterioration in sleep quality. What is required from an auto-adjusting device, therefore, is a reactive process to the types of respiratory event that warrant an increase or decrease in CPAP pressure in a timely and gradual manner.

The device must be able to recognise and react in a clinically effective manner to obstructive hypopnoea and obstructive apnoea. Whilst there are accepted clinical definitions for these two variants of obstructive events, revising those definitions for use in laboratory testing is not a straightforward process.

In 2012 the American Academy of Sleep Medicine (AASM) changed the definition of apnoea in its manual for the scoring of sleep and associated events [19] to “drop in the peak signal excursion by $\geq 90\%$ of pre-event baseline using an oronasal thermal sensor (diagnostic study), PAP device flow (titration study), or an *alternative* apnea sensor, for ≥ 10 seconds”.

The AASM definition of hypopnoea in its scoring manual [19] was changed to “scored when the peak signal excursions drop by $\geq 30\%$ of pre-event baseline using nasal pressure (diagnostic study), PAP device flow (titration study), or an *alternative* sensor, for ≥ 10 seconds in association with either $\geq 3\%$ arterial oxygen desaturation or an

arousal”.

In August 2013, AASM revised the definition of hypopnoea again^[20] so that the 2012 definition became rule 1a and was “Recommended”. The definition prior to 2012 was re-adopted as rule 1b: “scored when the peak signal excursions drop by $\geq 30\%$ of pre-event baseline using nasal pressure (diagnostic study), PAP device flow (titration study), or an *alternative* sensor, for ≥ 10 seconds in association with either $\geq 4\%$ arterial oxygen desaturation or an arousal”. Rule 1b was re-adopted as “Acceptable”.

EN ISO 80601-2-70:2020 refers to the AASM definition of obstructive hypopnoea as “diagnostic hypopnoea” and goes on to state that “**equipment hypopnoea** is diagnostic hypopnoea as estimated by sleep breathing therapy equipment”. *EN ISO 80601-2-70:2020* notes that “Sleep apnoea breathing therapy equipment without oximetry cannot monitor oxygen saturation. Therefore, sleep apnoea breathing therapy equipment without the oxygen saturation information, can only provide an estimate of hypopnoea events”.

The precise definition of equipment hypopnoea is left to be defined by the individual manufacturer. This variance in definition means that some leeway must be incorporated into a laboratory test.

10.2 Device Reaction to Recognised Respiratory Events

10.2.1 Obstructive Hypopnoea and Obstructive Apnoea - The device must react to obstructive events, both hypopnoea and apnoea of increasing or decreasing severity, adjusting Positive Airway Pressure (PAP) up or down as appropriate. The device needs to do so at a rate that achieves the required goal in the shortest possible time, but without sacrificing sleep quality. This requires an appropriate, gradual, change in PAP.

10.2.2 Central Hypopnoea and Central Apnoea - *EN ISO 80601-2-70:2020* does not require an APAP device to recognise central hypopnoea and central apnoea. What this means in clinical practice is that some APAP devices may react to central hypopnoea and central apnoea events by increasing CPAP pressure and other devices may not.

The whole purpose of laboratory testing whether a device reacts or not, therefore, is to give information to the clinician as to how a device behaves in varying clinical circumstances. A test of this functionality should only be performed on a device where the manufacturer claims that the device data download provides an estimate of central hypopnoea and central apnoea or that manufacturer claims that a device will recognise but not react by increasing pressure in response to central hypopnoea and apnoea events.

10.2.3 Periodic/Cheyne-Stokes Breathing - *EN ISO 80601-2-70:2020* does not require an APAP device to recognise Periodic/Cheyne-Stokes breathing. As with central apnoea and central hypopnoea, this means, in clinical practice, that some APAP devices may react to Periodic/Cheyne-Stokes breathing events by increasing CPAP pressure, and other devices may not.

A test of this functionality should only be performed on a device where the

manufacturer claims that the device data download provides an estimate of Periodic/Cheyne-Stokes breathing, or that manufacturer claims that a device will recognise but not react by increasing pressure in response to Periodic/Cheyne-Stokes breathing events.

10.3 Validation of APAP Devices

The capability of a device to recognise and react to respiratory events will be ascertained by independent laboratory testing according to the protocol specified in Appendix D.

DESIRABLE FEATURES

11.0 DESIRABLE FEATURES OF CPAP DEVICES

11.1 Auto On/Off Facility

There may be an Auto On/Off switching function. This is breath actuated (typically by 2 -3 breaths with the mask applied). This is a prescriber setting.

11.2 Auto-Adjusting Device Reporting Facilities

Auto-Adjusting devices must provide a download facility with appropriate software that produces a report showing graphical representations of (i) the delivered pressure versus time, (ii) any leak versus time and (iii) detection of any obstructive events (apnoeas, hypopnoeas, flow limitation and snore). The use of mean, median and 95th-centile pressure value (or similar) must be displayed. However, all suggested titration values should be clinically validated.

The accuracy of reporting of obstructive apnoea (Apnoea Index - AI) and hypopnoea (Apnoea-Hypopnoea Index - AHI) events from downloaded data must agree within 15% as compared to the indices determined by manual analysis of a Level 2 Sleep Study (Limited Channel Cardio-Pulmonary Sleep Study - Polygraphy) or Level 3/4 Sleep Study (Home Polysomnography or Standard Attended Polysomnography) recording over the same period of time.

No current international standard for the download format and structure of data from CPAP devices is in existence. It is the stated intention of ARTP SAC to set a standard for data download format and structure following consultation with CPAP manufacturers.

11.3 Software Features – All CPAP Devices

Special features (e.g., C-FLEX™, APAP, Pressure Relief, Leak v time, Detection of apnoea/hypopnoea events, etc.) should, wherever possible, be clinically validated in a peer-reviewed medical journal that has a reasonable citation index.

11.4 Data Storage/Exchange

Systems should support remote compliance uploads via a data card system or similar remote transfer technology allowing detailed usage to be viewed by support staff remotely located to the patient. Data storage must be for at least 6 months in the CPAP device, data card or through external storage (e.g., cloud technology).

The level of data stored should be deemed sufficient to make future clinical assessments. Software and appropriate hardware for data analysis for PC/Mac should be provided by the vendor to facilitate data analysis and record storage for each patient. Output of data from the software should be by both hard copy and PDF.

11.5 Internal Battery Backup

As the number of devices available with an internal battery backup capacity sufficient to enable functionality for several hours continues to increase, manufacturers must be able to validate claims of expected duration of full functionality without restoration of mains voltage.

The capability of a device to maintain full functionality under standard test conditions at a fixed therapy pressure of 10 hPa will be ascertained by independent laboratory testing according to the protocol specified in Appendix E.

11.6 Carry Bags

Devices should be supplied complete with a carry bag that will house all components and mask accessories. The maximum dimensions of the bag must not exceed 56 x 35 x 23 cm to comply with many international aviation specifications for carry on hand luggage.

12.0 REFERENCES

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TEST PROTOCOL

Determination of Noise Level in accordance with *EN ISO 80601-2-70:2020*

1.0 MEASUREMENT AND TEST EQUIPMENT

The measurement and test equipment used must be stated in the test report. Details of calibration of the sound level meter used must be stated in the test report.

2.0 MEASUREMENT CONDITIONS

Measurements must be made in a free sound field; i.e. a sound field region with no adjacent reflecting surfaces. In practice a free field exists if the *Direct* sound is 6 dB or preferably 10 dB greater than the *Reverberant* or reflected sound.

Direct sound is defined as the region in which the sound measured can be attributed to the source alone without the effects of obstructions, walls, other reflecting surfaces or adjacent noise sources.

Reverberant or reflected sound is defined as the region in a 'room' where the reflected sound dominates as opposed to the region close to the noise source where the direct sound dominates.

3.0 TEST SETUP

The test setup will be as set out as in Figure 1 below in accordance with the requirements of DIN ISO 3744:2011-02, EN ISO 3744:2010 (D).

DIN EN ISO 3744:2011-02
EN ISO 3744:2010 (D)

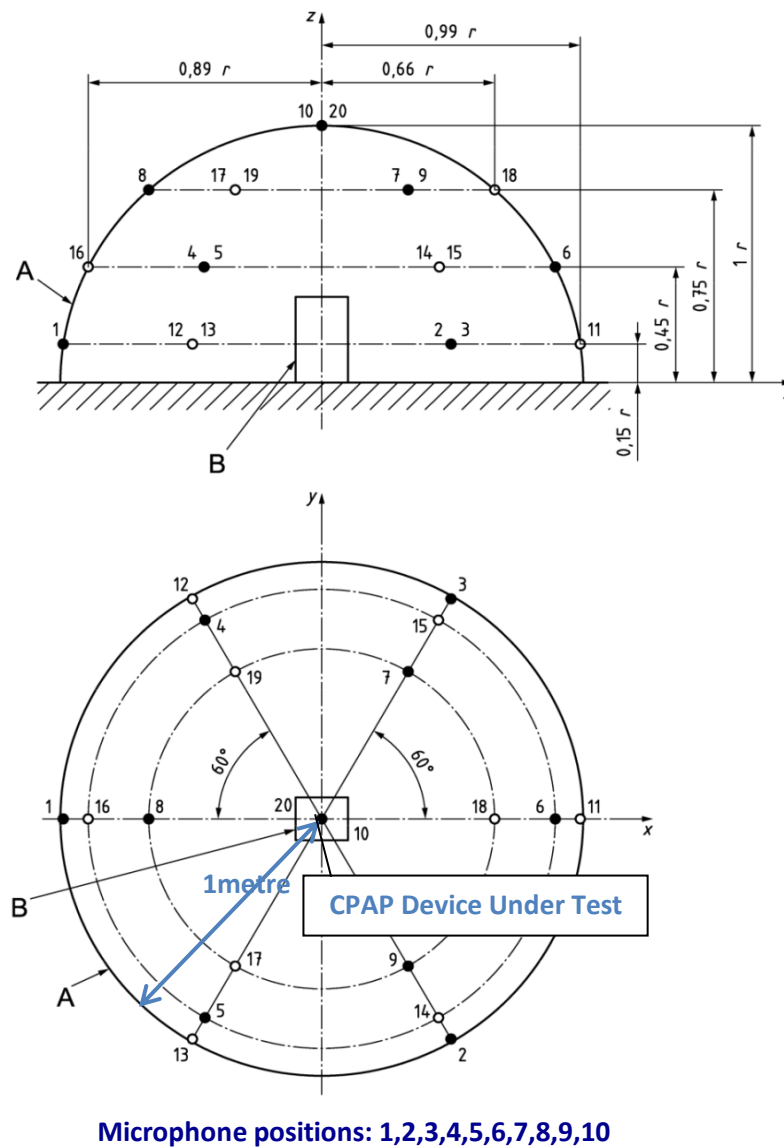


Figure 1: Test Setup (source: ISO 3744:2010)

4.0 TEST SEQUENCE

- 1) The CPAP device under test should be placed on a reflective plane according to *ISO 3744:2010*. More than one breathing circuit can be supplied by the manufacturer (different internal diameters and/or heated/non-heated delivery tube). The test shall be conducted with a breathing circuit specified by the manufacturer in instructions supplied with delivery of the device to be tested.
- 2) Connect the patient connection to the standard resistance.
- 3) The tubing system and the standard resistance must be acoustically isolated.
- 4) Measure the background noise and record it using the frequency-weighted characteristic A and the time-evaluated characteristic F.
- 5) Set the CPAP device under test to a pressure of 10 hPa and switch it on.
- 6) Measure the sound pressure level and record it using the frequency-weighted characteristic A and the time-evaluated characteristic F at the measuring points shown in Figure 1 above.

5.0 TEST RESULTS

5.1 Background Noise

The background noise measured during the test must be more than 6dB lower than the measured sound pressure level and preferably more than 10db lower.

5.2 Documentation of Sound Pressure Levels

The background noise and device sound pressure level from each of the 10 measuring points as shown in Figure 1 above should be documented in the test report in tabular format as shown in Table (a1) below. The table shows example test results.

Measuring points	Device noise ($L'_{p(ST)}$)	Background noise ($L_{p(ST)}$)
	db(A)	db(A)
1	25.5	16.4
2	25.7	16.4
3	27.9	16.4
4	29.3	16.4
5	27.5	16.4
6	28.6	16.4
7	30.9	16.4
8	27.5	16.4
9	28.0	16.4
10	30.6	16.4
$\overline{L'_{p(ST)}}$	28.48	16.4

Table (a1): Test results tabulation using example test results

5.3 Calculation of Mean Value of the Sound Pressure Levels

The mean value of the sound pressure levels should be calculated using the methodology below and the result determined documented in the bottom row of the table shown in Table (a1) above. The horizontal bar above the parameters $\overline{L'_{p(ST)}}$ and $\overline{L_{p(ST)}}$ is the mathematically correct format to indicate that the parameter is a mean value.

$$\overline{L'_{p(ST)}} = 10 \log \left[\frac{1}{N_M} \sum_{n=1}^{N_M} 10^{0.1 L'_{pn(ST)}} \right] \text{ dB(A)}$$

where:

$L'_{p(ST)}$: A-weighted sound pressure level, expressed in decibels, at the n^{th} microphone position or the n^{th} microphone positions, with the sound source (ST) being tested in operation

N_M : The number of microphone positions

To illustrate the calculation more clearly for non-mathematicians, the remainder of the calculation is shown using the example data in Table (a1) above:

$$\overline{L'_{p(ST)}} = 28.48 \text{ dB(A)}$$

5.4 Calculation of the Background Noise Correction Factor K_1

The background noise correction factor, K_1 , should be calculated as follows:

$$K_1 = -10 \log(1 - 10^{-0.1 \Delta L_p})$$

where:

$$\Delta L_p = \overline{L'_{p(ST)}} - \overline{L_{p(ST)}}$$

$\overline{L'_{p(ST)}}$: The mean value of the time-averaged A-weighted sound pressure levels, expressed in decibels, over all microphone positions on the measuring surface, while the sound source being tested in operation

$\overline{L_{p(ST)}}$: The calculated mean of the time-averaged A-weighted sound pressure levels of the background noise in decibels over all microphone positions on the measuring surface

Taking the example test results from Table (a) above:

$$\Delta L_p = \overline{28.48 \text{ dB(A)}} - \overline{16.4 \text{ dB(A)}}$$

$$\Delta L_p = 12.08 \text{ dB(A)}$$

$$K_1 = -10 \log(1 - 10^{-0.1 * 12.08 \text{ dB(A)}})$$

$$K_1 = 0,28 \text{ dB(A)}.$$

5.4 Calculation of the Environmental Correction Factor K_2 from the Absorption Area of the Test Room

The environmental correction factor, K_2 , should be calculated as follows:

$$K_2 = 10 \log \left(1 + 4 \frac{S}{A} \right) dB(A)$$

where:

A is defined as the equivalent sound absorption area of the room in square metres.

S is defined as the area of the measuring room in square metres.

$$A = \alpha * S_V$$

$$S = 2 \pi r^2$$

where:

α : the mean sound absorption rate given for A-weighted values in Table A.1 ISO 3744

S_V : the total area of the space-limiting areas of the measuring room (walls, ceiling and floor) in square metres

Taking some example data:

$\alpha = 0.5$, according to table A.1 in ISO 3744 (room with a lot of sound-absorbing material both on the ceiling and on the walls).

$$S_V = 150 \text{ m}^2$$

$$A = 0,5 * 150 \text{ m}^2$$

$$A = 75 \text{ m}^2$$

$$S = 2 \pi (1 \text{ m})^2 \text{ (Test area has a radius of 1 metre)}$$

$$S = 6.28 \text{ m}^2$$

$$K_2 = 10 \lg \left(1 + 4 \frac{6.28 \text{ m}^2}{75 \text{ m}^2} \right) dB(A)$$

$$\mathbf{K_2 = 1.25 dB(A).}$$

5.5 Calculation of the Corrected Sound Pressure Level

The corrected sound pressure level $\overline{L'_{p(ST)corr}}$ is calculated using the formula:

$$\overline{L'_{p(ST)corr}} = \overline{L'_{p(ST)}} - K_1 - K_2$$

Using the example data from 5.3, 5.4 and 5.5:

$$\overline{L'_{p(ST)corr}} = 28.48 \text{ dB(A)} - 0.28 \text{ dB(A)} - 1.25 \text{ dB(A)}$$

$$\overline{L'_{p(ST)corr}} = 26.95 \text{ dB(A)}.$$

The A - weighted sound pressure level produced by the CPAP device has a mean corrected value of 26.95 dB (A) after deduction of the appropriate correction factors.

6.0 TEST REPORT

The test report must state:

- a) The testing laboratory
- b) Who the test was conducted by
- c) The author of the test report
- d) The test device including serial number and firmware version
- e) The client for whom the test report has been commissioned
- f) The period over which the device was tested

The test report must include details of:

- i. The measurement and test equipment used
 - ii. The measurement conditions
 - iii. The test setup
 - iv. The test sequence
 - v. The test results as specified in Section 5 above
 - vi. The measurement uncertainty of the test method should be stated in the report as a percentage of the measured value
-

TEST PROTOCOL

Determination of Pressure Stability in accordance with *EN ISO 80601-2-70:2020*

1.0 MEASUREMENT AND TEST EQUIPMENT

The measurement and test equipment used must be stated in the test report.

2.0 MEASUREMENT CONDITIONS

The ambient conditions under which the test protocol is conducted must be stated in the test report:

- Temperature
- Barometric Pressure
- Relative Humidity

Any derogation from the stated test protocol must be documented in the test report.

3.0 TEST SETUP

The CPAP device under test should be set up according to the manual supplied by the manufacturer.

For an Auto-Adjusting CPAP device, the Fixed Pressure mode of operation should be selected.

More than one breathing circuit can be supplied by the manufacturer (different internal diameters and/or heated/non-heated delivery tube). The test shall be conducted with a breathing circuit specified by the manufacturer in instructions supplied with delivery of the device to be tested.

The CPAP device should be connected to the standard resistance using the breathing circuit supplied by the manufacturer as shown in Figure 2 on the following page.

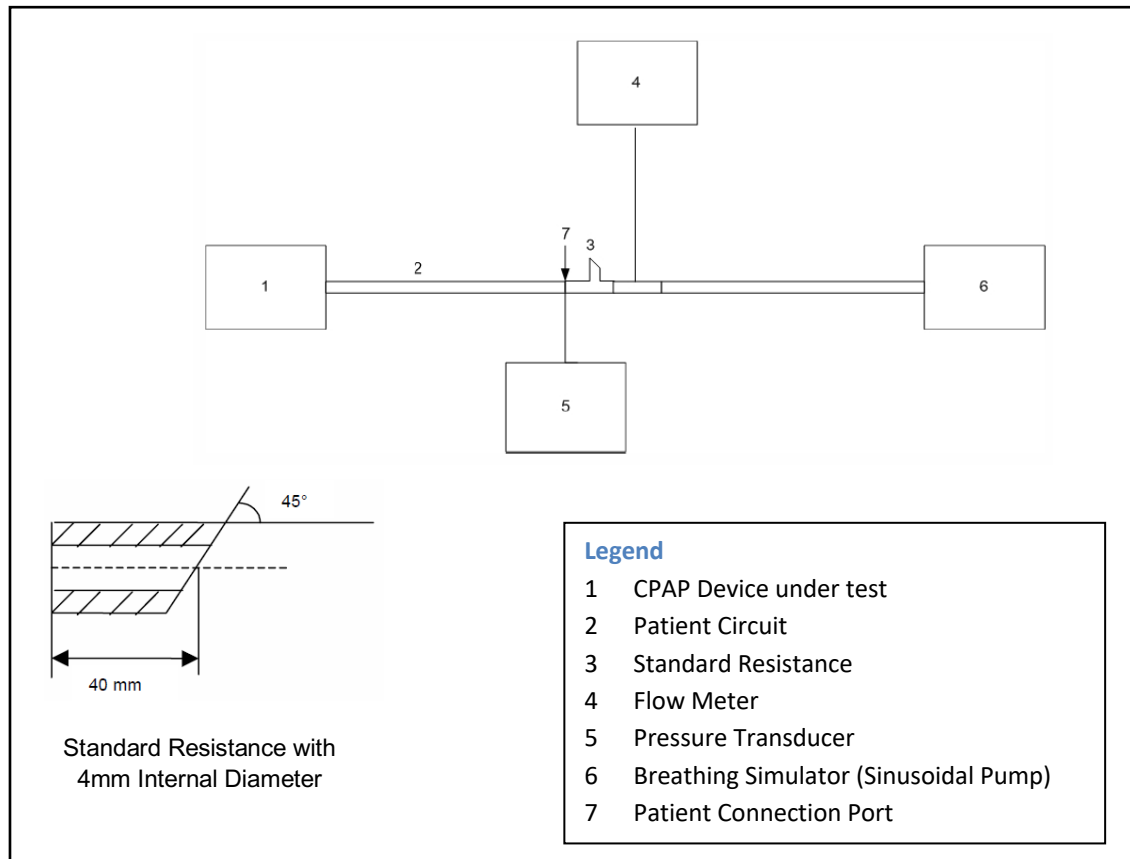


Figure 2: Test Setup (source ISO 80601-2-70:2015)

4.0 TEST SEQUENCE

- 1) The pressure difference between the inspiratory and expiratory phases, (δp , pressure difference), should be recorded in accordance with *ISO 80601-2-70:2020*.
- 2) The breathing simulator should be set to deliver a tidal volume (V_t) of 500 ml
- 3) The CPAP Pressure should be set at 5 pressures from the minimum pressure that can be set, p_{min} , to the maximum pressure that can be set, p_{max} , determined as specified in Table (b1) below.
- 4) At each of the 5 pressure levels recordings of the pressure waveform will be made for a period of at least 5 minutes. The Mean Pressure, $\bar{p}$, and the Pressure Difference, δp , will be determined at each test pressure level and breathing frequency.
- 5) A representative sample of the pressure waveform will be generated at each pressure level and breathing frequency and documented as *Waveform 1 to n* (to be presented as Figures 1 to n in an appendix to the report) as shown in Table (b2) below.

Pressure and flow must be recorded for each test sequence with a sampling frequency of at least 20 Hz. *EN ISO 80601-2-70:2020* states "At the patient connection port, flow

and pressure are monitored and measured with a measurement device for these parameters using a low-pass filter at 5 Hz and a minimum acquisition rate of 100 Hz.” From the point of view of measurement technology and signal processing, the specified 100Hz sampling rate is unnecessarily high, since the high-frequency signals have been filtered out. Therefore, 20 Hz is sufficient.

Set ventilation pressure p (hPa)	Breath frequency f (breaths/min)	Tidal volume V_t (ml)
p_{min}	10, 15, 20	500
$p_{min} + \frac{1}{4}(p_{max} - p_{min})$		
$p_{min} + \frac{1}{2}(p_{max} - p_{min})$		
$p_{min} + \frac{3}{4}(p_{max} - p_{min})$		
p_{max}		

Table (b1): Test Parameters

For example, if p_{min} is 4 hPa and p_{max} 20 hPa, then $\frac{1}{4}(p_{max} - p_{min}) = 4\text{hPa}$. The five pressure levels (p) as in Table (b1) above would be 4 hPa, 8 hPa, 12 hPa, 16 hPa and 20 hPa.

5.0 TEST RESULTS

The Mean Pressure, $\bar{p}$, and the Pressure Difference (Short Term Accuracy), δp , should be reported as in Table (b2) below.

Tidal Volume V_t (ml)	Set Ventilation Pressure p (hPa)	Breathing Frequency f (breaths/min)	Short-term Accuracy δp (hPa)	Mean Pressure $\bar{p}$ (hPa)	Figure in Appendix
500		10			1
		15			2
		20			3
		10			4
		15			5
		20			6
		10			7
		15			8
		20			9
		10			10
		15			11
		20			12
		10			13
		15			14
		20			15

Table (b2): Pressure Stability (Short Term Accuracy)

The measurement uncertainty of the test method should be stated in the report as a percentage of the measured value.

The recordings of the pressure waveform at each test pressure level and breathing frequency should be attached to the report as an appendix with each level being clearly labelled as in Figure 3 shown on the following page.

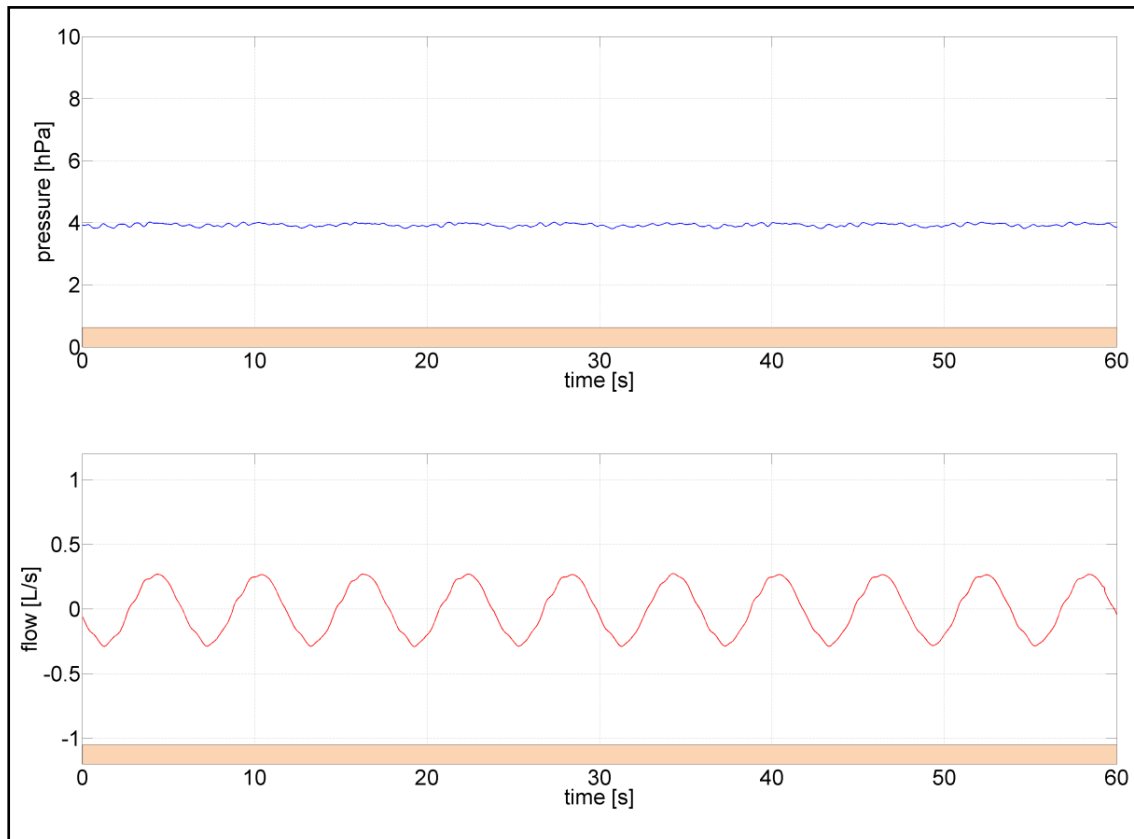


Figure n - Set Pressure: 4hPa
10 breaths per minute

Figure 3: Example Pressure Waveform Display as an Appendix to the Test Report

6.0 TEST REPORT

The test report must state:

- a) The testing laboratory
- b) Who the test was conducted by
- c) The author of the test report
- d) The test device including serial number and firmware version
- e) The client for whom the test report has been commissioned
- f) The period over which the device was tested

The test report must include details of:

- i. The measurement and test equipment used
- ii. The measurement conditions
- iii. The test setup
- iv. The test sequence

- v. The test results as specified in Section 5 above
 - vi. The measurement uncertainty of the test method should be stated in the report as a percentage of the measured value
-

TEST PROTOCOL

Determination of Relative Humidity in accordance with *EN ISO 80601-2-70:2020*

1.0 MEASUREMENT AND TEST EQUIPMENT

The measurement and test equipment used must be stated in the test report.

2.0 MEASUREMENT CONDITIONS

The ambient conditions under which the test protocol is conducted must be stated in the test report:

- Temperature
- Barometric Pressure
- Relative Humidity

The reservoir of the humidifier should be half filled. The water in the humidifier should be left to adjust to room temperature before commencing the test.

Any derogation from the stated test protocol must be documented in the test report.

3.0 TEST SETUP

The CPAP device under test should be set up according to the manual supplied by the manufacturer with the humidifier connected. More than one breathing circuit can be supplied by the manufacturer (different internal diameters and/or heated/non-heated delivery tube). The test shall be conducted with the breathing circuit specified by the manufacturer in instructions supplied with delivery of the device to be tested.

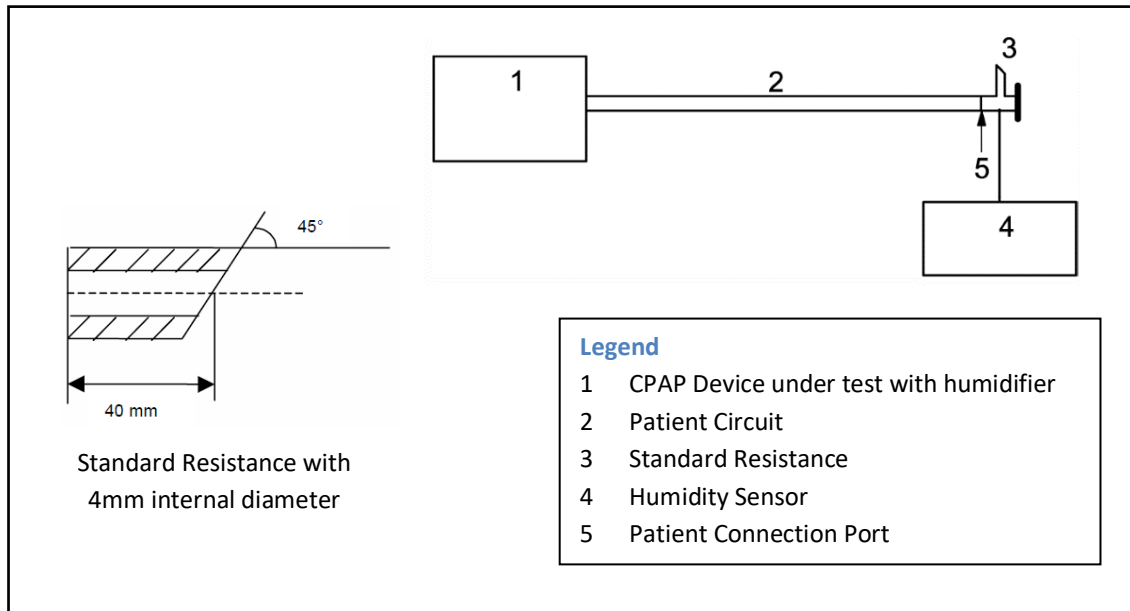


Figure 4: Test Setup

4.0 TEST SEQUENCE

The CPAP device should be turned on and pressure set to 10 hPa. For an Auto-Adjusting CPAP device, the operational mode should be set to Fixed Pressure and set to 10 hPa.

Relative Humidity and Temperature must be recorded for each test sequence with a sampling frequency of at least 1 Hz.

4.1 Standard (Non-Heated) Patient Circuit

- 1) If the device has a standard, non-heated patient circuit, it is to be assumed that the humidifier operation does not include an "auto" mode and the humidification level should be set to the lowest position.
- 2) Data acquisition should commence and continue throughout the entire measurement process.
- 3) When the relative humidity achieves a plateau, the humidifier control should be switched to the next position.
- 4) The sequence should continue until all humidification level steps have been completed.

4.2 Heated Patient Circuit

Depending upon the device under test, there may be more than one mode of operation. Specifically, some devices will have several levels of humidification set manually and some devices will have a fully automatic mode in addition. In this latter case, both modalities will need to be tested.

If the humidification is set manually, then the test sequence steps set in 4.1 above should be followed. If the device has a fully automatic mode, the following test sequence should be applied:

- 1) The humidifier 'automatic' mode should be activated.
- 2) Data acquisition should commence and continue until the relative humidity achieves a plateau for a duration of at least 15 minutes.

5.0 TEST RESULTS

The maximum relative humidity achieved together with the maximum temperature measured at the patient connection port should be reported. Graphical presentation of the entire data acquisition sequence showing Relative Humidity and Temperature measured at the patient connection port versus Time should be presented with the legend identifying both measured parameters. For manual mode testing, the humidification control step change positions should be identified on the graph.

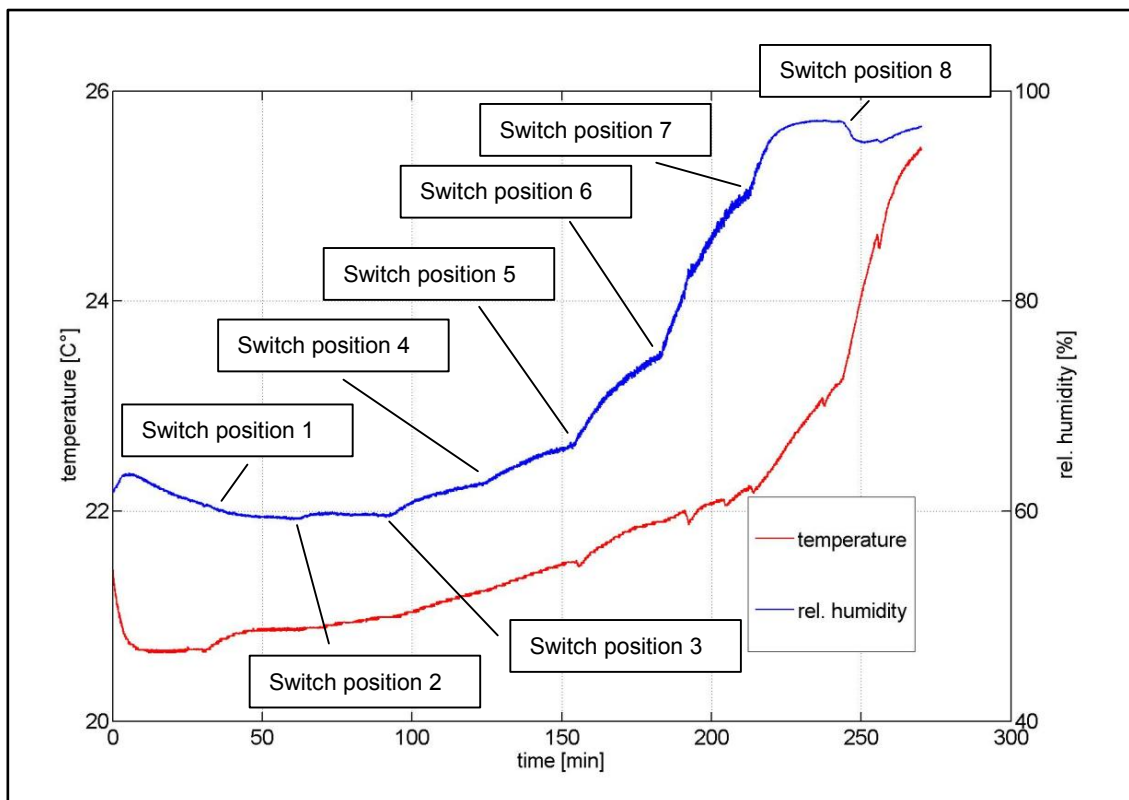


Figure 5: Graphical Presentation of Results for Manual Mode Humidification Testing Showing Steps

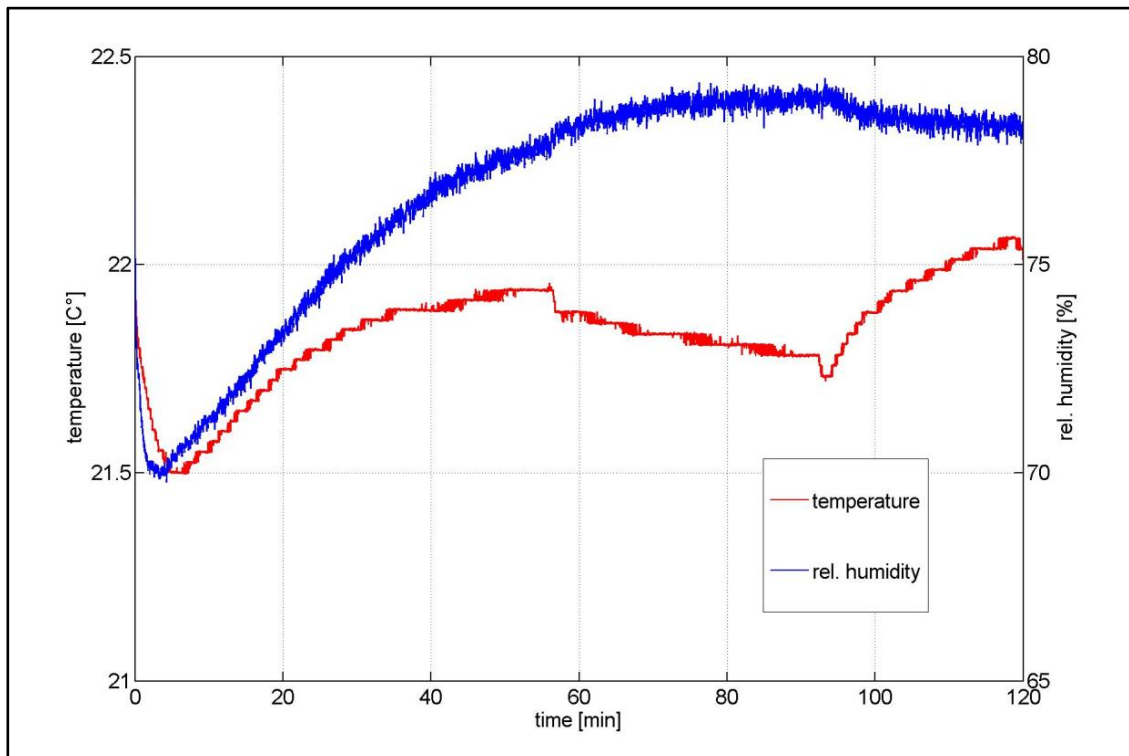


Figure 6: Graphical Presentation of Results for Automatic Mode Humidification Testing Showing End Plateau

6.0 TEST REPORT

The test report must state:

- a) The testing laboratory
- b) Who the test was conducted by
- c) The author of the test report
- d) The test device including serial number and firmware version
- e) The client for whom the test report has been commissioned
- f) The period over which the device was tested

The test report must include details of:

- i. The measurement and test equipment used
- ii. The measurement conditions
- iii. The test setup
- iv. The test sequence
- v. The test results as specified in Section 5 above
- vi. The measurement uncertainty of the test method should be stated in the report as a percentage of the measured value

DRAFT TEST PROTOCOL

Determination of Auto-Adjusting CPAP Device detection of, and response to, respiratory events

1.0 MEASUREMENT AND TEST EQUIPMENT

The measurement and test equipment used must be stated in the test report.

2.0 MEASUREMENT CONDITIONS

The ambient conditions under which the test protocol is conducted must be stated in the test report:

- Temperature
- Barometric Pressure
- Relative Humidity

Any derogation from the stated test protocol must be documented in the test report.

3.0 TEST SETUP

The APAP device under test should be set up according to the manual supplied by the manufacturer.

APAP devices may have one or more modalities of operation in auto-adjusting mode. For example, there are now commercial devices available which refer to gender specific modes of auto-adjustment. Furthermore, more than one breathing circuit can be supplied by the manufacturer (different internal diameters and/or heated/non-heated delivery tube). The test shall be conducted with a breathing circuit and auto-adjusting mode (for all genders) specified by the manufacturer in instructions supplied with delivery of the device to be tested.

The APAP device (1) should be set up according to the manufacturer's instruction manual connected to a pressure transducer (4), a standard resistance (5), a flow meter (6), an obstruction simulator (7) with pressure generator (8) and a lung simulator (9) using the breathing circuit (2) supplied by the manufacturer as shown in Figure 7 on the following page.

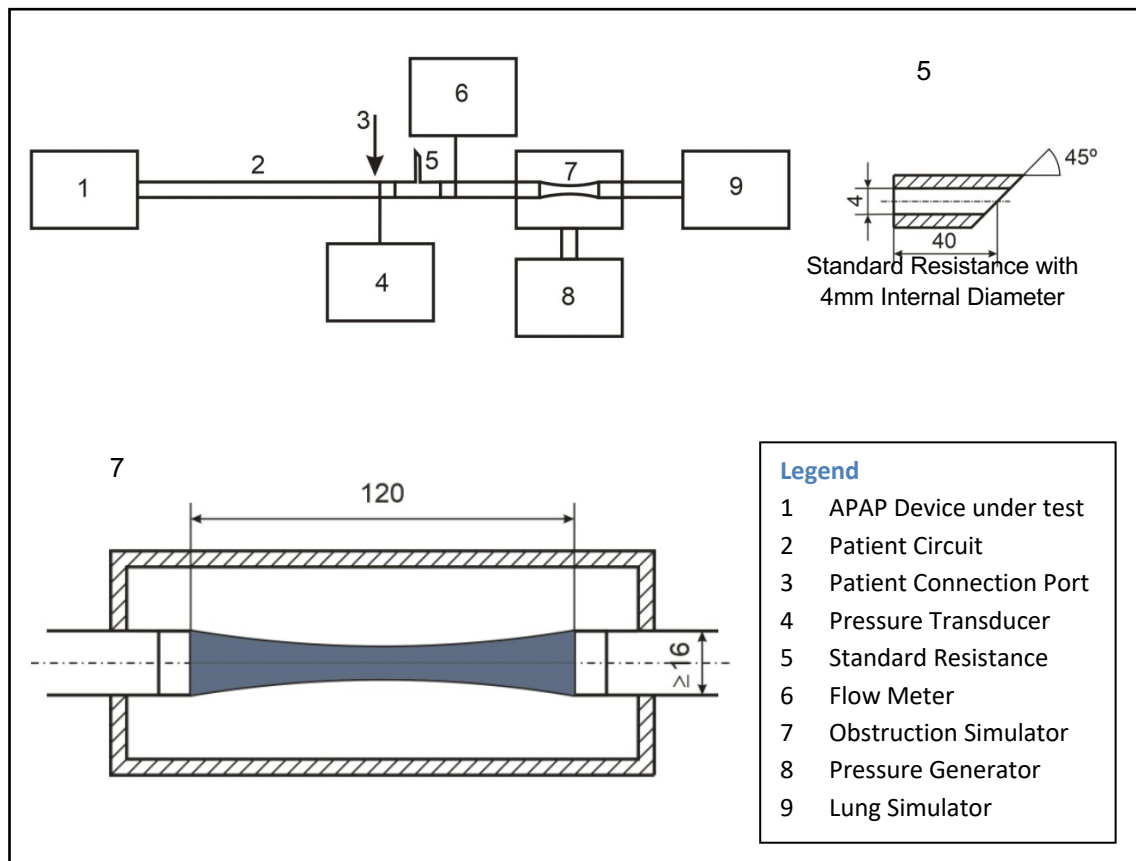


Figure 7: Test Setup

4.0 TEST SEQUENCES

Test sequences are introduced into an event-free breathing simulation using a lung and obstruction simulator with a breath volume of 500 ml, a ratio of inspiration to expiration I/E = 1:2 and a respiratory frequency at 14 breaths per minute. Obstructive and central apnoea and hypopnoea events are simulated as are periodic breathing and Cheyne-Stokes breathing.

The lower pressure limit of the APAP device must be set to 4 hPa and the upper pressure limit must be set to 20 hPa so that the device can regulate the PAP within these limits. Before each test sequence, the APAP device must be restarted to avoid influences of any previous test sequence on automatic pressure adjustment. Each test sequence will commence with at least 20 minutes of normal breathing. Measurement data will be collected and evaluated for each test sequence.

A series of simulated individual respiratory events will be used to perform initially five and ultimately six scenarios used to test the basic automatic control behaviour of the auto-adjusting device. Pressure and flow must be recorded for each test sequence with a sampling frequency of at least 20 Hz.

Obstructive Apnoea is simulated by complete occlusion of a collapsible element. This is achieved by applying a pressure of 10 hPa from the pressure generator (8) to the exterior of the collapsible element. The collapsible element is fully opened at a CPAP therapy pressure of 10 hPa.

Obstructive Hypopnoea is the most difficult event to simulate in a laboratory test setting. Obstructive hypopnoea is accompanied by noise arising from tissue vibration associated with upper airway collapse. Most APAP devices recognise obstructive hypopnoea by detecting reduction in airflow associated with noise arising from physical vibration from within the patient circuit. It is important therefore that any simulation of obstructive apnoea introduces realistic noise into the simulation. The noise, however, must be generated as a result of physical vibration associated with airway collapse and not simply by introducing noise using a loudspeaker or similar auditory generator. This test simulation is currently in the development stage and is therefore not included for the present time.

Central Apnoea is simulated by simply pausing the breathing simulator.

Central Hypopnea is simulated by reducing the simulated respiratory volume output from the breathing simulator by 75%.

Periodic Breathing is defined for the purposes of this test protocol as clusters of normal breathing separated by intervals of hypopnoea which is central in nature. It is simulated by a sequence of normal breathing with regular periods of hypopnoea.

Cheyne-Stokes Breathing is an abnormal pattern of breathing characterized by progressively deeper and sometimes faster breathing, followed by a gradual decrease that results in a central apnoea. The pattern repeats, with each cycle usually taking between 30 seconds and 2 minutes.

4.1 Test 1 - Detection of Obstructive Apnoea

The test simulation comprises 3 sequences after at least 20 minutes of normal breathing for each sequence. Each sequence consists of 20 successive obstruction phases (10 hPa) with a duration of 15 seconds (Sequence 1), 30 seconds (Sequence 2) and 60 seconds (Sequence 3). There is a phase of normal breathing of 60 seconds duration between each obstruction in each sequence. The sequences are depicted in Figure 8.

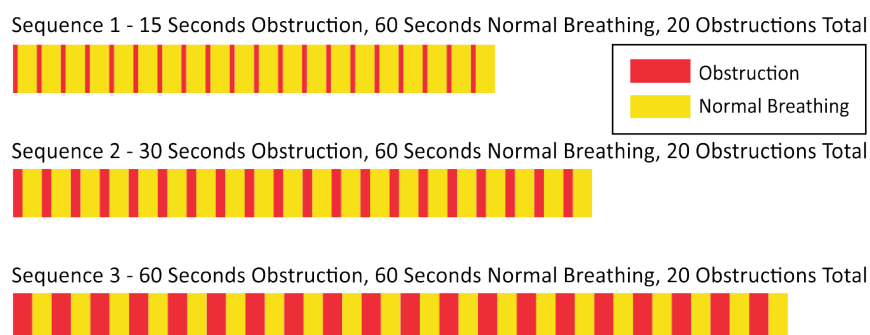


Figure 8: Test 1 Simulation Sequences

Required Outcomes for Test 1

For each sequence the device must respond by commencing to increase PAP by at least 0.3 hPa in no more than 5 obstruction phases.

4.2 Test 2 - Response to Obstructive Apnoea

The test simulation comprises a sequence of 50 obstruction phases (10 hPa) after 20 minutes of normal breathing comprising 10 'batches' of 5 obstructions with each obstruction having a duration of 15 seconds followed by a 20 second phase of normal breathing. There will be a 60 second period of normal breathing between each of the obstruction batches. The sequence is depicted in Figure 9.

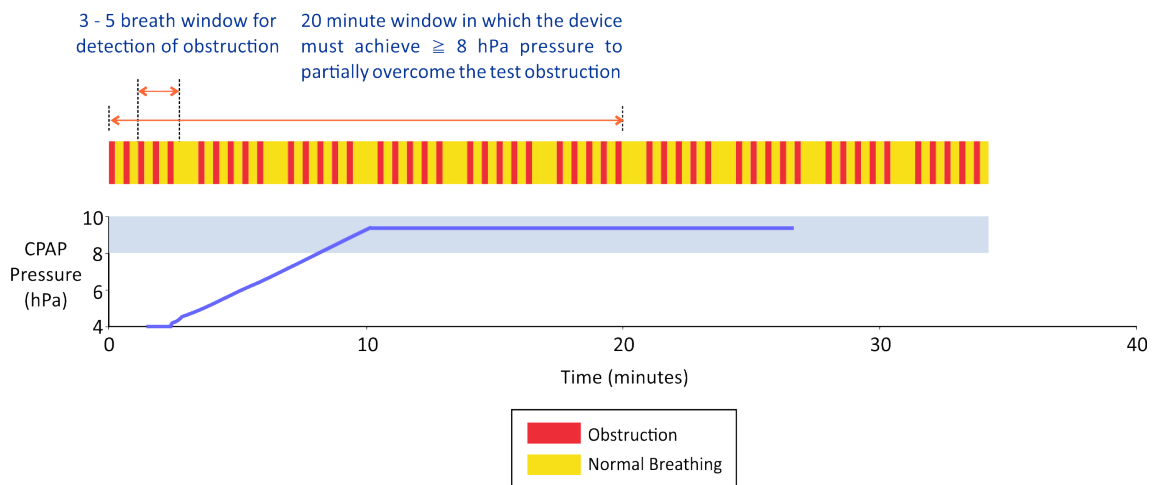


Figure 9: Test 2 Simulation Sequence

Required Outcome for Test 2

The device must achieve a pressure of at least 8 hPa to partially overcome the test obstruction in no more than 20 minutes.

4.3 Test 3 - Detection of Obstructive Hypopnoea

In Development

4.4 Test 4 - Response to Obstructive Hypopnoea

In Development

4.5 Test 5 - Detection and Response to Central Apnoea

The test simulation comprises two sequences:

- i) **Sequence 1** commences after 20 minutes of normal breathing and comprises 10 phases of 60 seconds central apnoea followed by 60 seconds of normal breathing. Sequence 1 is depicted in Figure 10.
- ii) **Sequence 2** commences after 20 minutes of normal breathing and comprises 20 phases of 20 seconds central apnoea followed by 60 seconds of normal breathing. Sequence 2 is depicted in Figure 10.

The simulation waveform as apnoea occurs and recovers is depicted in Figure 11.

Sequence 1 - 60 Seconds Central Apnoea, 60 Seconds Normal Breathing, 10 Sequences Total



Sequence 2 - 20 Seconds Central Apnoea, 60 Seconds Normal Breathing, 20 Sequences Total

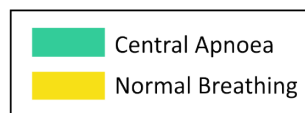


Figure 10: Test 5 Simulation Sequences

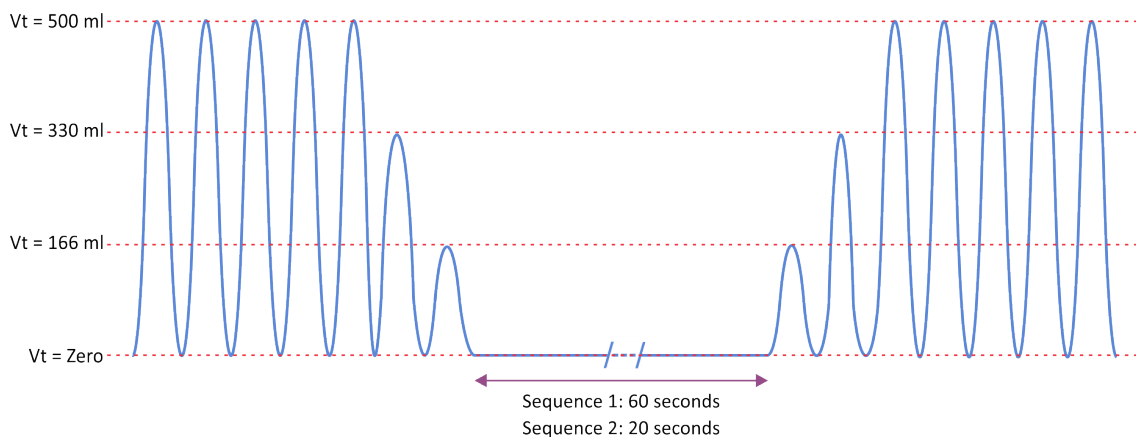


Figure 11: Central Apnoea Simulation Waveform

Required Outcomes for Test 5

The device must recognise but not alter PAP in response to Sequences 1 and 2. Recognition is defined for this test as not increasing PAP to > 5 hPa.

4.6 Test 6 - Detection and Response to Central Hypopnoea

Central hypopnoea is simulated by a 75% reduction in Tidal Volume. For each event, the Tidal Volume reduces from 100% in a two-breath sequence to 75% followed by 50% to the target level of 25% to achieve the 75% reduction. The reverse occurs at the end of the hypopnoea duration. The test simulation comprises two sequences:

- i) **Sequence 1** commences after 20 minutes of normal breathing and comprises 10 phases of 60 seconds central hypopnoea followed by 60 seconds of normal breathing. Sequence 1 is depicted in Figure 12.
- ii) **Sequence 2** commences after 20 minutes of normal breathing and comprises 20 phases of 20 seconds central hypopnoea followed by 60 seconds of normal breathing. Sequence 2 is depicted in Figure 12.

The simulation waveform as hypopnoea occurs and recovers is depicted in Figure 13.

Sequence 1 - 60 Seconds Central Hypopnoea, 60 Seconds Normal Breathing, 10 Sequences Total



Sequence 2 - 20 Seconds Central Hypopnoea, 60 Seconds Normal Breathing, 20 Sequences Total

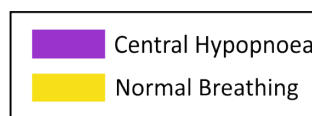


Figure 12: Test 6 Simulation Sequences

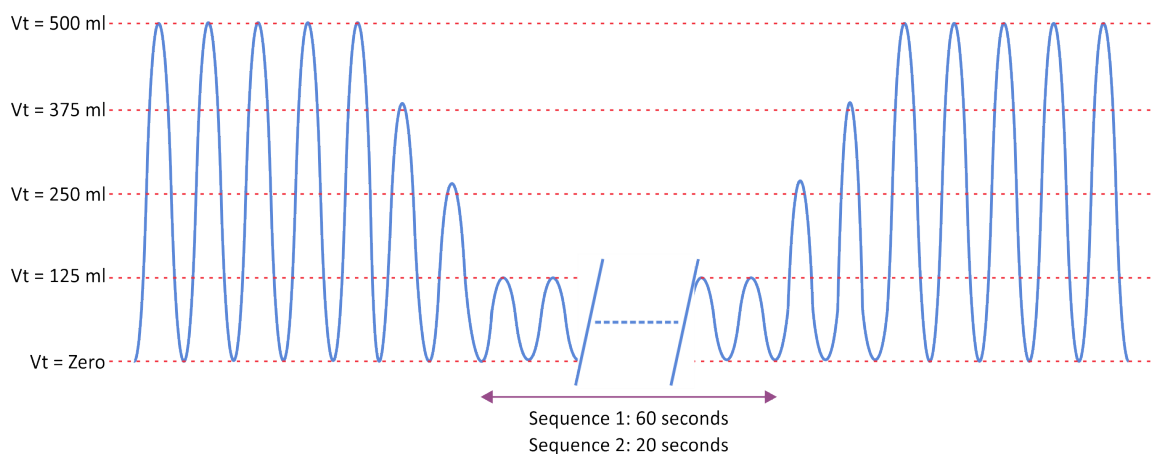


Figure 13: Central Hypopnoea Simulation Waveform

Required Outcomes for Test 6

The device must recognise but not alter PAP in response to Sequences 1 and 2. Recognition is defined for this test as not increasing PAP to > 5 hPa.

4.7 Test 7 - Detection and Response to Periodic Breathing

The sequence commences after 20 minutes of normal breathing with Tidal Volume set at 500 ml throughout this period. Over 7 breaths, the Tidal Volume is decreased by 250 ml to 250 ml. Over the following 7 breaths, the Tidal Volume is increased by 250 ml to 500 ml. The cycle is repeated a further 9 times to make 10 cycles in total. The sequence is depicted in Figure 14.

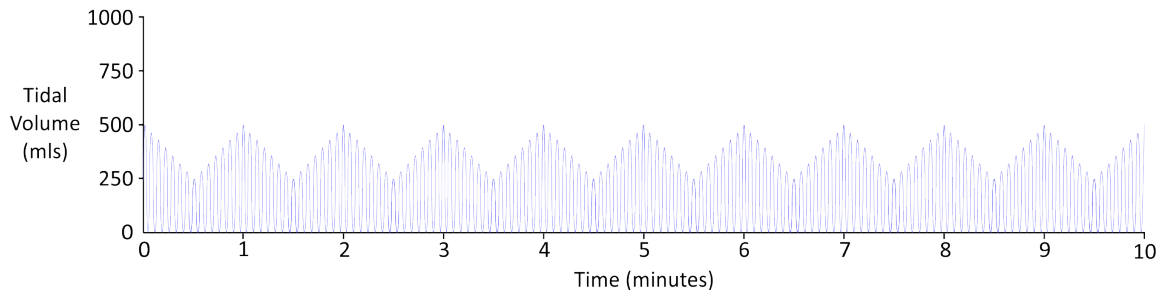


Figure 14: Test 7 Simulation Sequence

Required Outcome for Test 7

The device must recognise but not alter PAP in response to the sequence. Recognition is defined for this test as not increasing PAP to > 5 hPa.

4.8 Test 8 - Detection and Response to Cheyne-Stokes Breathing

The sequence commences after 20 minutes of normal breathing with Tidal Volume set at 500 ml throughout this period. Over 3 breaths, the Tidal Volume is increased by 250 ml to 750 ml. Over the following 7 breaths, the Tidal Volume is reduced to 500 ml to 250 ml. A 20 second central apnoea ensues. Over the following 7 breaths the Tidal Volume is increased by 500 ml to 750 ml. The cycle is repeated a further 9 times to make 10 cycles in total. The sequence is depicted in Figure 15.

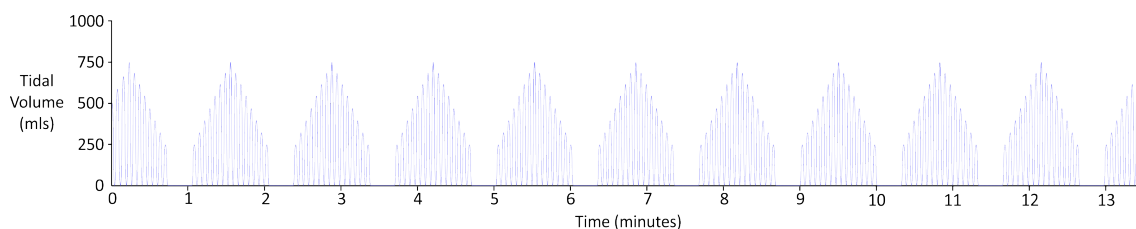


Figure 15: Test 8 Simulation Sequence

Required Outcome for Test 8

The device must recognise but not alter PAP in response to the sequence. Recognition is defined for this test as not increasing PAP to > 5 hPa.

5.0 TEST RESULTS

The outcome of each test performed (Tests 1 through 8) should be documented. The measured pressure and flow patterns must be graphically displayed over time for all tests and sequences based on the recorded raw data.

The recorded raw data for pressure shall be filtered forward and backward with a 3rd order Butterworth low pass filter with a cut-off frequency of 0.1 Hz. Based on the filtered pressure values and the flow values, the following information must be determined for all sequences of tests and presented in tabular form:

Test 1 - Detection of Obstructive Apnoea

	15s obstruction	30s obstruction	60s obstruction
Obstruction detected and treatment commenced?	Y/N	Y/N	Y/N
Number of obstruction phases required to detect and commence treatment of obstruction			

Test 2 - Response to Obstructive Apnoea

Response Criterion	
≥ 8 hPa Therapeutic Pressure achieved?	Y/N
Time to achieve ≥ 8 hPa Therapeutic Pressure (mm:ss)	
Highest achieved Therapy Pressure (hPa)	

Test 3 - Detection of Obstructive Hypopnoea

In Development

Test 4 - Response to Obstructive Hypopnoea

In Development

Tests 5 and 6 - Detection and Response to Central Apnoea and Central Hypopnoea

Sequence 1 120 Event /120 seconds Normal Breathing		Central Apnoea	Central Hypopnoea
Increase in Therapy Pressure to > 5 hPa observed?		Y/N	Y/N
Highest Therapy Pressure Achieved	Value (hPa)		
	After how many phases?		

Sequence 2 20 Event /60 seconds Normal Breathing		Central Apnoea	Central Hypopnoea
Increase in Therapy Pressure to > 5 hPa observed?		Y/N	Y/N
Highest Therapy Pressure Achieved	Value (hPa)		
	After how many phases?		

Tests 7 and 8 - Detection and Response to Cheyne Stokes and Periodic Breathing

		Cheyne Stokes	Periodic Breathing
Increase in Therapy Pressure to > 5 hPa observed?		Y/N	Y/N
Highest Therapy Pressure Achieved	Value (hPa)		
	After how many phases?		

6.0 TEST REPORT

The test report must state:

- a) The testing laboratory
- b) Who the test was conducted by
- c) The author of the test report
- d) The test device including serial number and firmware version
- e) The client for whom the test report has been commissioned
- f) The period over which the device was tested

The test report must include details of:

- i. The measurement and test equipment used
 - ii. The measurement conditions
 - iii. The test setup
 - iv. The test sequences
 - v. The test results as specified in Section 5 above
 - vi. The measurement uncertainty of the test method should be stated in the report as a percentage of the measured value
-

7.0 VARIATION OR ADJUSTMENT OF TESTS AND TEST WAVEFORMS

It is always possible that a device manufacturer will design a device with operational algorithms which, in their opinion, functions effectively, with robust evidence of this from clinical trials. ARTP would not wish to disadvantage any such manufacturer.

Should any such situation arise, the onus is on the manufacturer wishing to submit such a device for independent evaluation through the ARTP programme to draw this to the attention of ARTP under guarantee of absolute confidentiality, including independent testing facilities which may be engaged from time to time by ARTP.

ARTP and its engaged independent testing facilities undertake, where good clinical evidence is provided, to work with device manufacturers to produce amended or alternative test waveforms in order to maintain fair and equitable practice.

DRAFT TEST PROTOCOL

Determination of Internal Battery Backup Duration

1.0 MEASUREMENT AND TEST EQUIPMENT

The measurement and test equipment used must be stated in the test report.

2.0 MEASUREMENT CONDITIONS

The ambient conditions under which the test protocol is conducted must be stated in the test report:

- Temperature
- Barometric Pressure
- Relative Humidity

Any derogation from the stated test protocol must be documented in the test report.

3.0 TEST SETUP

The device under test should be set up according to the manual supplied by the manufacturer.

Auto-adjusting devices will be tested in fixed pressure mode.

More than one breathing circuit can be supplied by the manufacturer (different internal diameters and/or heated/non-heated delivery tube). The test shall be conducted with a breathing circuit and auto-adjusting mode (for all genders) specified by the manufacturer in instructions supplied with delivery of the device to be tested.

The device will be connected to mains power for at least 12 hours immediately prior to the test to ensure 100% battery charge.

The device will be set to deliver a CPAP pressure of 10 hPa under *ISO 80601-2-70:2020* standard conditions as set out in Figure 16 on the following page.

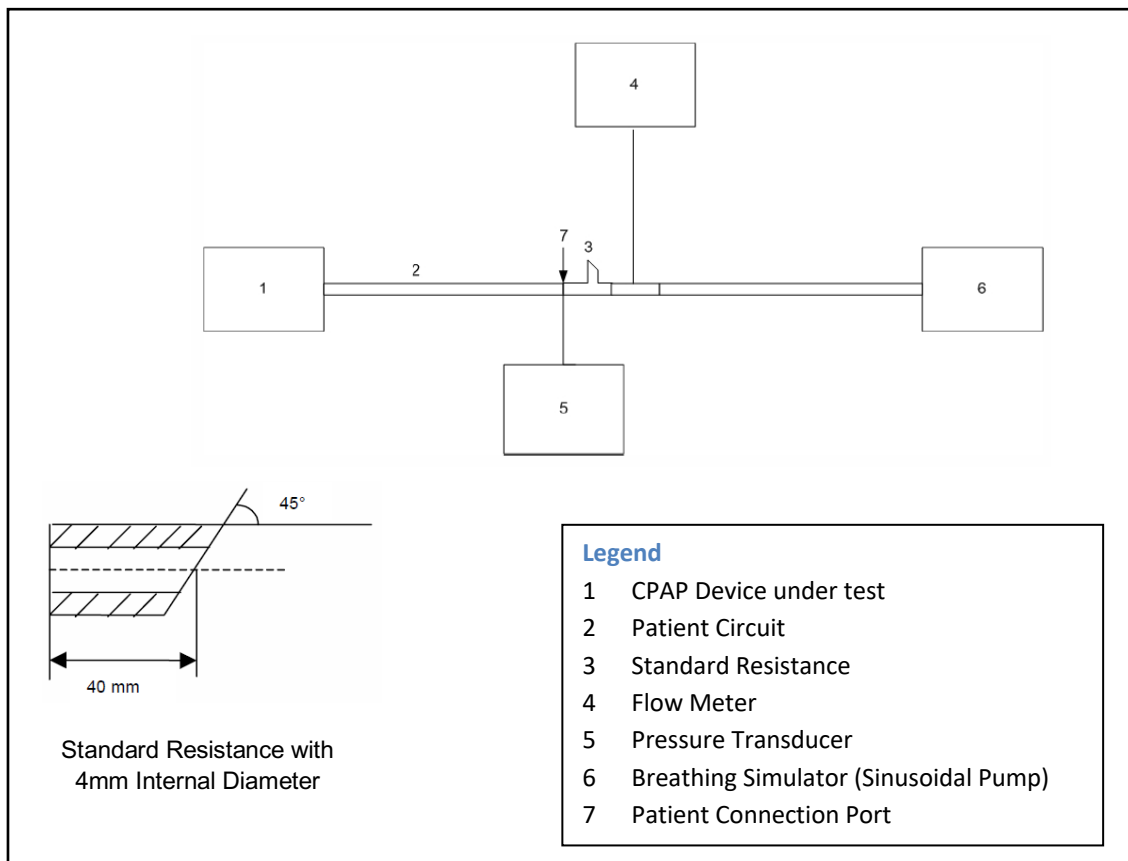


Figure 16: Test Setup (source ISO 80601-2-70:2015)

4.0 TEST SEQUENCE

The test sequence will be introduced into an event-free breathing simulation using a lung simulator with a breath volume of 500 ml, a ratio of inspiration to expiration I/E = 1:2 and a respiratory frequency at 14 breaths per minute.

Pressure and flow must be recorded for each test sequence with a sampling frequency of at least 20 Hz.

Required Outcome for the Test

The device must maintain a CPAP pressure of 10 hPa $\pm$ 0.5 hPa for the time specified by the manufacturer as being the claimed performance duration under standard test conditions with a fully charged internal battery.

5.0 TEST RESULTS

The outcome of the test should be documented. The measured pressure and flow patterns must be graphically displayed over time based on the recorded raw data.

The recorded raw data for pressure shall be filtered forward and backward with a 3rd order Butterworth low pass filter with a cut-off frequency of 0.1 Hz.

6.0 TEST REPORT

The test report must state:

- a) The testing laboratory
- b) Who the test was conducted by
- c) The author of the test report
- d) The test device including serial number and firmware version
- e) The client for whom the test report has been commissioned
- f) The period over which the device was tested

The test report must include details of:

- i. The measurement and test equipment used
 - ii. The measurement conditions
 - iii. The test setup
 - iv. The test sequence
 - v. The test results as specified in Section 5 above
 - vi. The measurement uncertainty of the test method should be stated in the report as a percentage of the measured value
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